

Nutritional Status, Infant

CCG: C-1157 (CCG)

MCG Health
Chronic Care
28th Edition

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Assessment

Q1. Caregiver: How much does your baby weigh and how long (tall) is your baby?

- Weight: _____
- Height: _____
- Unknown [P1] [P2]

Q2. Caregiver: What do you feed your baby? (Check all that apply.)

- Breast milk [Q4] [Q5]
- Formula, specify type: _____ [Q4] [Q5]
- Enteral nutrition (tube feeding) [Q6] [P2]
- Parenteral nutrition [Q6]
- Other, specify: _____ [Q4] [Q5] [P2]
- Nothing [B1] [P1] [P2]
- Unknown [Q4] [Q5] [B1] [P1] [P2]

Q3. Caregiver: How do you know that your baby is ready to eat? (Check all that apply.)

- Cries
- Fusses
- Grimaces
- Hand to mouth
- Rooting
- Sucking
- Other, specify: _____ [Q4] [Q5] [P2]
- Unknown [Q4] [Q5] [B1] [P1] [P2]

Q4. Caregiver: How often do you feed your baby?

- 6 times a day or less [B1] [P1] [P2]
- 7 to 12 times a day
- 13 times a day or more [P1] [P2]
- Nothing by mouth (gets alternative nutrition via enteral or parenteral nutrition) [P2]
- Other, specify: _____ [P1] [P2]
- Unknown [P1] [P2]

Q5. Caregiver: How long does your baby feed?

- 20 minutes or less [B1] [P1] [P2]
- 21 to 45 minutes
- 46 minutes or more [B1] [P1] [P2]
- Other, specify: _____ [P1] [P2]
- Unknown [P1] [P2]

Q6. Caregiver: What is your baby's nutrition schedule?

- Type, specify: _____
- Frequency, specify: _____
- Other, specify: _____
- Nothing [B1] [P1] [P2]
- Unknown [B1] [P1] [P2]

Q7. Caregiver: Is your baby gaining weight as expected?

- Yes
- No [P1] Growth and Development, Pediatric  CCG
- Unknown [P1] Growth and Development, Pediatric  CCG

Q8. Caregiver: Do you have any concerns about your baby's nutrition? (Check all that apply.)

- Affording [B1] [P2]
- Baby seems still hungry after just eating. [P2]
- Baby's behavior (eg, cries a lot, arches back, pulls away) [B1] [P2]
- Baby's suck (eg, does not latch on, breathing problems while feeding) [P2]
- Breast milk collection, storage, or use [P2]
- Breast milk production (eg, not enough, breast problems) [P2]
- Formula preparation [P2]
- Medicine taken by baby [P2]
- Medicine taken by breast-feeding mother [P2]
- Positioning baby (eg, problems holding or sitting up) [P2]
- Timing of feeding [P2]
- Other, specify: _____ [P1] [P2]
- No
- Unknown [B1] [P1] [P2]

Plan of Care

Barriers

B1. Nutritional deficit for infant

Problems

P1. Healthy diet information needed for infant

- **Goal:** Infant's nutritional status optimized
 - **Interventions:**
 - **ASSIST:** Identifying caregiver's goal priorities, willingness/ability to achieve goals, and preferences for communication and follow-up
 - **ASSIST:** Goal setting that is specific, measurable, achievable, relevant, and with target completion time (eg, short/long-term, prioritized)
 - **EDUCATE:** Healthy body weight and height for infant based on age:
 - Normal infant growth depends on gestational age.(1)
 - Growth rate of full-term infants after 10 days of life is generally 20 to 30 grams/kg per day.(1)
 - Weight gain for full-term infants is generally 680 grams per month.(2)
 - Growth rate for preterm infants is from 10 to 30 grams/kg per day.(3)
 - **EDUCATE:** Nutritional intake provision, with healthy, safe eating habits(2)(4)(5):
 - Breast milk or formula for first 12 months
 - Infants usually eat 8 to 12 times per day, with 2 to 4 hours between feedings.(6)
 - Feed infant on cue (eg, rooting, pre-cry grimacing, fussing, crying). Crying is a late cue and may make feeding harder.
 - Avoid giving infant cow's milk, low-iron milk (eg, goat, soy), and infant cereal.[A]
 - Expect increase of feeding during growth spurts.
 - **EDUCATE:** Feeding techniques(2)(4):
 - Hold baby upright and close to the body when feeding.
 - Burp infant frequently.
 - Manage choking episodes.
 - Calm baby before feeding (eg, dim lights, soothing touch, rocking, swaddling).
 - Talk to infant while feeding.
 - Do not prop bottle or enlarge bottle hole.
 - Cleaning, storage, heating, and care of bottles and nipples
 - **EDUCATE:** Monitor infant's output(4):
 - Keep diary of oral intake and urine, stool, and emesis output.
 - Count diapers and note results (eg, urine/stool, color, amount).
 - Factors impacting stool frequency: type of nutrition (eg, breast milk, formula type) and age
 - Check for problems (eg, more or less than usual, changes colors)
 - **EDUCATE:** Solid food introduction(5):
 - 4 to 6 months: usually ready developmentally

- Signs infant may be ready for solids: tongue thrust reflex (pushing food out of mouth) fades, coordinated swallowing, sits with support, and good head and neck control
- 6 to 12 months: usually master chewing and swallowing, able to handle finger foods, use cups and spoons, and try new tastes and textures
- Begin with one single-ingredient new food at a time; do not introduce new foods for 3 to 5 days to observe for allergy.
- Iron-fortified infant rice cereal least likely to cause allergy[B]
- Introduce soft moist foods in a variety of flavors and textures.
- It may take up to 15 attempts before infant accepts a particular food.
- **EDUCATE: Oral care(5):**
 - Clean gums with plain water twice a day using clean moist washcloth or small-headed soft-bristled toothbrush.
 - First tooth and all teeth when they erupt (6 to 10 months): brush with fluoridated toothpaste twice a day (after breakfast and before bed).
 - Dental caries (tooth decay) causes (eg, frequent/prolonged feedings, intake of high-sugar sources (eg, fruit drinks, soft drinks, fruit juice))
 - Avoid feeding at night to prevent sugary fluids from pooling around the teeth to help prevent dental caries (tooth decay).
 - Do not put infant to sleep with a bottle or sippy cup.
 - Oral health of parent impacts infant's oral health: parents can transmit caries-promoting bacteria to their infant; oral health care for parent (eg, dental care, see dentist regularly, limit dietary sugar).
- **EDUCATE: Feeding problem care; check and adjust(5):**
 - Feeding position (should be semi-erect, comfortable)
 - Hunger cue responsiveness
 - Environment; remove or relocate to limit distractions.
 - Colic signs (eg, cries inconsolably for hours, passes a lot of gas); offer more-frequent feedings.
- **EDUCATE: Resources:**
 - Women, Infants, and Children: www.fns.usda.gov/wic
 - U.S. National Library of Medicine: www.nlm.nih.gov
- **EDUCATE: When to get emergency help:**
 - Respiratory impairment (eg, gasping for air, choking, cyanosis, severe dyspnea)(7)
 - Dehydration (eg, breathing hard, weight loss, eyes sunken, fontanelles sunken, irritability, not responding to sounds or sights)(8)(9)
 - Aspiration (eg, coughing, choking, hypoxia, wheezing)
- **EDUCATE: When to call the provider:**
 - Crying more for no reason or crying that cannot be soothed
 - Feeding problems
 - Urine problems (eg, frequency change, blood)
 - Bowel problems (eg, frequency change, blood)
 - Infection (eg, low-grade fever, vomiting, diarrhea)
 - Caregiver cannot take care of baby anymore.
- **COORDINATE: Dietitian referral, if indicated(10)**
- **COORDINATE: Home care referral, if indicated**
- **COORDINATE: Community services referral, as needed[C](11)**
- **COORDINATE: Social worker referral, as needed**
- **COORDINATE: Speech-language pathologist referral, if indicated**
- **COORDINATE: Swallowing evaluation referral, if indicated**
- **COORDINATE: Laboratory testing, if indicated**
- **COORDINATE: Care team meeting, if needed(12)(13)**
- **COORDINATE: Communication between care team members and care plan reconciliation(12)(14)(15)**
- **COORDINATE: Follow-up to evaluate infant's nutritional status, referral status, and need for further assistance(16)(17)**
- **SEND: How to breast feed  (Spanish)**
- **SEND: Intake and output diary (Illustrated)  (Spanish)**
- **Goal: Breast-feeding information understood(4)**
 - **Interventions:**
 - **EDUCATE: Breast-feeding frequency:**
 - First 2 weeks after birth: 10 to 12 times a day
 - After first 2 weeks: 8 to 12 times a day for the next few months
 - After 4 months: 6 to 12 times a day
 - **EDUCATE: Breast-feeding duration: recommended for 6 months exclusively, and can continue after 12 months**
 - **EDUCATE: Breast-feeding techniques(5):**
 - Time needed for breast to let down
 - Allow infant to finish one breast before offering other breast.

- Early weeks of life: feed 10 to 12 times a day and wake infant to try feeding if asleep for more than 4 hours.
- Length of feeding should not be restricted (20 to 45 minutes is generally enough).
- Frequent feeding establishes milk supply and keeps breasts from getting too full.
- Avoid artificial nipples (eg, pacifiers) until breast-feeding established (usually by 4 weeks).(4)
- **EDUCATE:** Breast care and problem symptoms (eg, infection signs)
- **EDUCATE:** Collecting and storing breast milk when unable to direct feed; time limits of frozen breast milk
- **EDUCATE:** Breast milk supplementation, as needed(18):
 - First week of life: 400 International Units (IU) of vitamin D per day(4)
 - Iron 1 mL/kg per day when food is introduced (around 4 months)
 - Weaned infants can stop supplements when they get at least 1 quart (1 L) per day of vitamin D-fortified formula or whole milk.[A](5)
- **EDUCATE:** Mother's diet's effects on breast milk(5):
 - Drink a glass of water (fluoridated) at every feeding; drink whenever thirsty.
 - Limit caffeine (eg, coffee, tea, soft drinks) to 2 servings a day.
 - Avoid drugs and alcohol.
 - Possible maternal vitamin/mineral supplementation (eg, B12, iodine) if deficient (eg, vegan diet, chronic illness)(2)
- **EDUCATE:** Infant expected urine and stool frequency(2)(5)(19):
 - Expect 6 to 8 wet diapers and 3 or 4 stools per day, and weight gain at appropriate rate.
 - Stool frequency may decrease; by 6 weeks, stools may occur every 3 days.
- **SEND:** How to breast feed  (Spanish)
- **Goal:** Formula feeding information understood(2)(4)(6)
 - **Interventions:**
 - **ASSIST:** Identifying food resource problems (eg, affording, accessibility)
 - **EDUCATE:** Iron-fortified formula is preferred, unless contraindicated.[A]
 - **EDUCATE:** Formula preparation:
 - Birth to 1 month: average 20 ounces (600 mL) of formula per day
 - Prepare and provide 2 ounces (60 mL) formula every 2 to 3 hours at first; provide more if infant still seems hungry.
 - After 1 month: 24 to 31 ounces (720 to 930 mL) of formula per day
 - Mix formula correctly (not lumpy or too watery).
 - Use tap water or a safe water source that contains fluoride to mix powdered formula.
 - Prepared formula (eg, not needing mix): cover and refrigerate after opening.
 - Powdered formula: store at room temperature.
 - Do not add foods (eg, cereal) to formula.
 - Discard leftover formula when infant done feeding (do not reuse bottle).
 - **EDUCATE:** Infant expected urine and stool frequency(19):
 - Voiding 10 to 20 times per day
 - Stool: 1 to 2 stools per day, or may stool only every other day

P2. Potential nutrition problems for infant

- **Goal:** Infant's nutritional problems resolved
 - **Interventions:**
 - **ASSIST:** Identifying caregiver's goal priorities, willingness/ability to achieve goals, and preferences for communication and follow-up
 - **ASSIST:** Goal setting that is specific, measurable, achievable, relevant, and with target completion time (eg, short/long-term, prioritized)
 - **ASSIST:** Appointment scheduling
 - **ASSIST:** Identifying food resource problems (eg, affording, accessibility)
 - **ASSIST:** Creation of nutrition diary
 - **ASSIST:** Assess for food or medication impacting nutritional status.
 - **EDUCATE:** Nutritional recommendations for infant's health status
 - **EDUCATE:** Food and drug interactions potentially impacting nutritional status
 - **EDUCATE:** Feeding problem care(20)(21):
 - Frequently try to feed; allow to feed as tolerated if interested in feeding.
 - Nasal congestion problems: instill nasal saline drops and suction nares with bulb syringe before feeding.
 - Oral rehydration solutions (eg, Infalyte or Pedialyte): use only if recommended by provider.
 - Notify provider if intake or output outside expected parameters.
 - **EDUCATE:** Monitor infant's output(4):
 - Keep diary of oral intake and urine, stool, and emesis output.
 - Count diapers and note results (eg, urine/stool, color, amount).
 - Check that infant gets enough fluids.
 - Check for problems (eg, more or less than usual, changes colors).
 - **COORDINATE:** Community services referral, as needed(11)

- **COORDINATE:** Developmental status assessment with appropriate government organizations (eg, county school system)(11)
- **COORDINATE:** Dietitian referral, if indicated(10)
- **COORDINATE:** Home care referral, if indicated
- **COORDINATE:** Pharmacy medication support, if indicated(5)
- **COORDINATE:** Protective services or legal authorities, as needed
- **COORDINATE:** Rehabilitation therapy referral, if indicated
- **COORDINATE:** Social worker referral, as needed
- **COORDINATE:** Specialist referral, if indicated
- **COORDINATE:** Speech-language pathologist referral, if indicated
- **COORDINATE:** Swallowing evaluation referral, if indicated(5)
- **COORDINATE:** Laboratory testing, if indicated
- **COORDINATE:** Care team meeting, if needed(12)(13)
- **COORDINATE:** Communication between care team members and care plan reconciliation(12)(14)(15)
- **COORDINATE:** Follow-up to evaluate infant's nutritional status, referral status, and need for further assistance(16)(17)
- **SEND:** How to breast feed  (Spanish)
- **SEND:** Intake and output diary (Illustrated)  (Spanish)
- **SEND:** Weight diary (Illustrated)  (Spanish)

Evidence Summary

Specialty society: The American Academy of Pediatrics (AAP) recommends exclusive breast-feeding for at least the first 4 months of life, ideally 6 months.(5)

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Footnotes

[A] Cow's milk should not be given to infants younger than 12 months.(5) [A in Context Link 1, 2, 3]

[B] Infants at high risk for developing allergies are those with at least one first-degree relative (ie, parent, sibling) with allergic disease. Breast-feeding exclusively for at least 4 months or using hydrolyzed formulas may prevent or delay the occurrence of atopic dermatitis, cow's milk allergy, and asthma in infancy and early childhood.(5) [B in Context Link 1]

[C] For infants with special needs, nutritional problems may be addressed with a nutritional program in an early intervention program.(5) [C in Context Link 1, 2]

MCG Health
Chronic Care 28th Edition
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Last Update: 3/14/2024 3:29:35 AM
Build Number: 28.1.20240313235330.016020